MINI PROJECT 3

AI-Based Cloud Security Threat Model

Cybersecurity Project Report

STUDENT DETAILS

Name:	Vukoti Reethusri
Roll / Registration No.:	STU-36527D
Batch:	Cyber Security/G17
Platform:	Sure proED
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AI-assisted analysis for safer cloud configurations

1. Abstract

The AI-Based Cloud Security Threat Model is a proposed system that helps identify and assess security risks in cloud-hosted applications and infrastructure. Cloud environments may contain storage buckets, databases, virtual machines, APIs, identity accounts, and network rules. Incorrect permissions or weak configurations can expose data or allow unauthorized access. The proposed system accepts a cloud architecture description or a safe configuration export, applies threat-modeling rules, and uses AI assistance to explain potential risks and recommend defensive controls. Findings are prioritized for human review. The system is intended as a decision-support tool and does not guarantee that a cloud environment is secure.

2. Introduction

Cloud computing lets organizations use computing power, storage, databases, and applications over a network. Security responsibilities are shared between the cloud provider and the customer, and the exact division depends on the service model. Customers must still configure identity permissions, data access, network exposure, logging, and encryption appropriately. Threat modeling provides a structured way to understand what needs protection, how an attacker might misuse a design, and which controls reduce risk.

3. Problem Statement

Cloud systems can become vulnerable through public data exposure, excessive permissions, insecure APIs, missing audit logs, or weak configuration. Teams may have difficulty reviewing many resources and understanding which issues matter most. This project proposes a tool that maps important cloud assets, identifies plausible threats from provided configuration data, ranks findings, and produces understandable mitigation recommendations.

4. Objectives

• Identify cloud assets, trust boundaries, and sensitive data flows from a supplied architecture or sample inventory.

• Recognize common cloud security risks using documented rules and a threat-modeling framework.

• Use AI assistance to explain possible impact and suggest practical defensive controls.

• Prioritize findings by exposure, asset importance, likelihood, and potential impact.

• Generate a report that can be reviewed and acted on by a human security analyst.

5. Scope and Example Architecture

A demonstration architecture may include a user, a web application, an API, a cloud identity service, a database, and object storage. Data flows between these components should be recorded. The model checks whether the design includes appropriate authentication, least-privilege permissions, restricted storage access, encryption, monitoring, and backup protection.

Component	Purpose	Security questions
Users / Identity service	Authenticates people and services	Is MFA enabled? Are permissions least-privilege?
Web app / API	Processes requests and data	Are requests authenticated and inputs validated?
Database	Stores application records	Is network access restricted? Is data encrypted?
Object storage	Stores files and backups	Can unauthorized users access objects?
Logging / monitoring	Records activity and alerts	Are important events logged and reviewed?

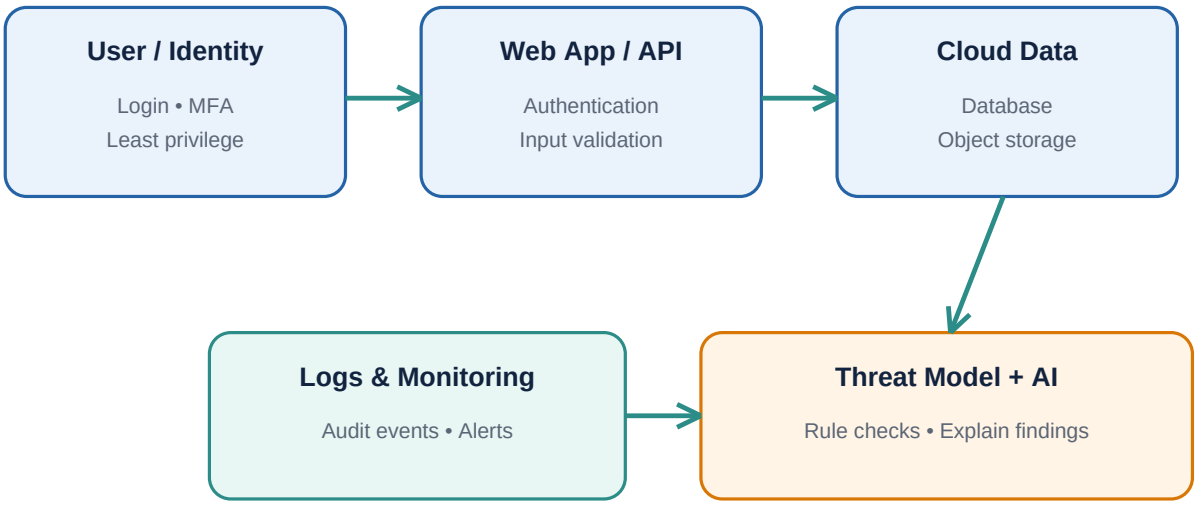
6. Proposed Workflow

Step 1 — Input: Provide an architecture diagram, manually entered asset inventory, or a sanitized configuration export from an account you are authorized to assess.

Cloud Security Architecture

Illustrative system diagram — not a live cloud scan

Example assets and data flow



Security checks represented

- MFA + least privilege
- Restricted network access
- Private storage + encryption
- Audit logs + human review

Use only authorized cloud assets and sanitized example configurations.

Step 2 — Parse and validate: Identify resources, permissions, data sensitivity, network exposure, and logging status. Flag missing fields instead of guessing.

Step 3 — Threat analysis: Apply threat-modeling categories such as STRIDE and cloud-specific security checks.

Step 4 — Risk prioritization: Score each finding using impact, exposure, asset criticality, and available evidence.

Step 5 — AI assistance: Produce a short explanation and remediation suggestion grounded in the observed configuration and trusted documentation.

Step 6 — Human review: Verify findings, approve changes, and record remediation status before making production changes.

7. Common Threat Scenarios

Threat	Possible impact	Recommended control
Publicly accessible storage	Sensitive files may be exposed	Block public access by default; review bucket and object policies
Excessive identity permissions	An account may access or change more resources than required	Apply least privilege; review roles and rotate credentials when appropriate
Unrestricted management ports	Services may be reachable from untrusted networks	Restrict ingress to approved networks; prefer managed access paths
Missing audit logging	Suspicious activity may go unnoticed	Enable relevant audit logs, protect them, and configure alerts
Unencrypted sensitive data	Data may be exposed if access or storage is compromised	Use appropriate encryption and secure key management
Insecure API access	Unauthorized requests may expose or alter records	Enforce authentication, authorization, validation, and rate limits

8. Role of AI

AI can help summarize architecture information, group related findings, explain why a configuration may be risky, and draft mitigation steps. A safer design is to combine deterministic security rules with AI explanations: rules identify measurable conditions, while AI makes results easier to understand. The system should provide evidence for each finding, cite trusted guidance where available, and ask for human review when the evidence is incomplete.

AI should not invent resources, assume a setting is unsafe without evidence, or automatically change cloud configurations. Treat configuration files and AI-generated text as potentially sensitive or untrusted input.

9. Illustrative Risk Prioritization

For a student prototype, a simple 0–100 score can make the ranking understandable. This formula is illustrative and is not an official cloud-provider standard.

Factor	Weight	Example
Potential impact	35%	Could the issue expose sensitive data or interrupt a critical service?
Exposure	30%	Is the resource reachable from the public internet or a broad network?
Asset criticality	20%	How important is the affected application or data?
Evidence / likelihood	15%	Is there verified evidence that the condition is exploitable or actively targeted?

Example formula: Score = (Impact ÷ 5 × 35) + (Exposure ÷ 5 × 30) + (Asset criticality ÷ 5 × 20) + (Evidence ÷ 5 × 15), where each factor is rated from 0 to 5.

Illustrative bands: 80–100 Critical; 60–79 High; 35–59 Medium; below 35 Low. Tune the bands to project policy. A confirmed urgent security advisory or severe exposure may require escalation regardless of the score.

10. Sample Findings

The following sample is fictional and demonstrates expected output only; it is not a scan of a real cloud account.

Finding	Evidence	Example priority	Suggested action
Storage allows public access	Sample policy indicates anonymous read access	Critical	Confirm business need; block public access if not required and review access logs
Admin role too broad	Role includes unnecessary write permissions	High	Reduce permissions to the minimum required and review recent activity
Audit logging disabled	Sample configuration says audit events are not recorded	High	Enable appropriate audit logs and alert on important events

11. Suggested Technology Stack

Technology	Purpose
Python	Threat rules, risk scoring, and report generation
Streamlit	Simple interactive dashboard
JSON / CSV	Store sanitized sample assets and configuration findings
Optional AI model	Summarize evidence and draft explanations; avoid sending secrets to external services

Sample Threat Assessment Dashboard

Fictional findings for demonstrating project output

3	Sample findings

1	Critical

2	High

Sample finding	Priority	Evidence	Suggested action
Storage allows public access	CRITICAL	Anonymous read	Review policy; block public access if not needed
Admin role too broad	HIGH	Extra write rights	Reduce to least-privilege permissions
Audit logging disabled	HIGH	Events not recorded	Enable audit logs and alerting

Illustrative risk score scale

Scores are mock examples, not measurements from a real cloud account.

12. Benefits

• Provides a structured way to review cloud design and configuration risks.

• Helps prioritize the issues that may have the greatest impact.

• Explains technical findings in simpler language for students and project reviewers.

• Encourages least privilege, secure storage, logging, encryption, and careful access management.

• Creates a repeatable report that can be used to track remediation.

13. Limitations and Safety

• The tool only assesses the data it receives; missing configuration details may hide risks.

• AI-generated findings can be wrong or incomplete and must be verified.

• Cloud security responsibilities differ by provider and service model.

• Do not include credentials, access tokens, private keys, personal data, or other secrets in sample files or AI prompts.

• Use only cloud accounts and configurations you own or have explicit permission to assess; review changes before applying them.

14. Future Enhancements

• Add provider-specific checks for AWS, Microsoft Azure, or Google Cloud.

• Import findings from authorized configuration-audit tools.

• Add a dashboard with severity filters and remediation status.

• Link recommendations to official provider documentation and track evidence.

• Evaluate the model against a labeled test set and measure missed risks and false alarms.

15. Conclusion

The AI-Based Cloud Security Threat Model offers a practical way to study how cloud assets, identities, data flows, and configuration choices can create security risks. By combining explicit checks with AI-assisted explanations, the system can make threat analysis easier to understand and help teams prioritize defensive improvements. Reliable evidence, data protection, provider-specific guidance, and human review remain essential.

16. References and Further Reading

• NIST SP 800-30 Rev. 1, Guide for Conducting Risk Assessments: https://csrc.nist.gov/pubs/sp/800/30/r1/final

• NIST SP 800-144, Guidelines on Security and Privacy in Public Cloud Computing: https://csrc.nist.gov/pubs/sp/800/144/final

• OWASP Top 10: https://owasp.org/www-project-top-ten/

• Microsoft Cloud Security Benchmark: https://learn.microsoft.com/security/benchmark/azure/

• AWS Well-Architected Framework, Security Pillar: https://docs.aws.amazon.com/wellarchitected/latest/security-pillar/welcome.html

• Google Cloud Security Foundations Guide: https://cloud.google.com/architecture/security-foundations